Chuanyang Jin

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Education

New York University Sep 2020 – Expected May 2024
B.A in Computer Science with Honors; B.A in Mathematics GPA: 3.96, Rank: 1/375

Publications

MMToM-QA: Multimodal Theory of Mind Question Answering
Chuanyang Jin, Yutong Wu, Jing Cao, Jiannan Xiang, Yen-Ling Kuo, Zhiting Hu, Tomer Ullman, Antonio Torralba, Joshua Tenenbaum, Tianmin Shu.
Under review (ACL, Nominated for Best Paper Award), 2024. **[Webpage]** **[Code]** **[ArXiv:2401.08743]**

Neural Amortized Inference for Nested Multi-agent Reasoning
Kunal Jha, Tuan Anh Le, Chuanyang Jin, Yen-Ling Kuo, Joshua Tenenbaum, Tianmin Shu.
AAAI, 2024. **[Webpage]** **[Code]** **[ArXiv:2308.11071]**

Dynamics of RNA Localization to Nuclear Speckles are Connected to Splicing Efficiency
Jinjun Wu*, Yu Xiao*, Yunzheng Liu*, Li Wen, Chuanyang Jin, Shun Liu, Sneha Paul, Chuan He, Oded Regev, Jingyi Fei.
Under review (Molecular Cell), 2024. **[BioRxiv:2024.02.29.581881]**

The Cultural Psychology of Large Language Models: Is ChatGPT a Holistic or Analytic Thinker?
Chuanyang Jin*, Songyang Zhang*, Tianmin Shu, Zhihan Cui.
Under review (PNAS), 2023. **[ArXiv:2308.14242]**

Self-Supervised Image Captioning with CLIP
Chuanyang Jin.
NeurIPS Self-Supervised Learning Workshop, 2023. **[ArXiv:2306.15111]**

Experiences

Massachusetts Institute of Technology Jun 2023 - Now
Research Intern – Supervisor: *Prof.* Joshua B. Tenenbaum and *Prof.* Tianmin Shu *Boston*

- Benchmark and propose a novel method for multimodal Theory of Mind capacities
  - Introduce MMToM-QA benchmark through procedural generation, which comprehensively evaluates the cognitive ability of multimodal models and LLMs to understand people's minds (e.g., beliefs, goals).
  - Propose a method BIP-ALM, which extracts and aligns symbolic representations from multimodal data, and integrates language-model-based reasoning with rule-based planning, achieving state-of-the-art results.
  - First-author paper submitted to ACL 2024.
- Accelerate nested multi-agent reasoning through neural amortized inference
  - Introduce a method that recursively performs particle sampling using recognition networks and leveraging data from generative models to amortize multi-agent reasoning and emulate human social inference.
  - Responsible for all theoretical proof and analyses. Paper published at AAAI 2024.

New York University Jun 2022 - Now
Undergraduate Research Intern – Supervisor: *Prof.* Oded Regev, *Prof.* He He and *Prof.* Saining Xie *New York*

- Use interpretable machine learning to derive insights into RNA localization processes
  - Analyze speckle-enriched transcripts and reveal sequence logics that drive RNA localization to nuclear speckles, utilizing interpretable machine learning to decipher complex structures and interactions.
  - Responsible for all computational aspects after biological experiments. Paper submitted to Molecular Cell.
- Improve the training and inference efficiency of Diffusion Transformers

- Implement optimizations such as pre-extracted VAE features, improved attention mechanisms, gradient checkpointing, and mixed precision training, achieving a 95% speed increase and 60% memory reduction.
- Integrate a pre-trained DistilBERT encoder to enable rapid and high-quality text embedding.
- Project received over 400 GitHub Stars, and was integrated into libraries and product implementations.

Chinasoft International Ltd. Jun 2021 – Aug 2021

Software Development Intern *Nanjing*

- Maintain and extend the functionalities of a company's personnel/financial data management web platform.
- Design, implement, and test tools using MySQL to enable administrators to operate on shared databases.
- Leverage Django web framework, JavaScript, and HTML/CSS to create a user-friendly interactive interface.

Projects

Self-Supervised Image Captioning with CLIP May 2023 - Jul 2023

- Propose a self-supervised captioning method by maximizing the CLIP relevance between images and captions.
- Leverage CLIP encoding, mapping network, and LLaMA 2. Surpass existing methods utilizing only 2% of data.
- First-author paper presented at NeurIPS'23 Self-Supervised Learning Workshop.

AutoCorp: A Fully Autonomous Company Apr 2023 - May 2023

- AutoCorp uses GPT-4 as a controller to generate reasoning traces and perform sequential actions. It generates t-shirt designs with DALL·E and tools, interacts with customers to sell products, and learns from feedback.
- Launch AutoCorp with startup co-founders as a pioneer product, gain over 800 users.

Selected awards

NYU Computer Science Prize for the Most Promising Junior Student (1 person/year)	2023
Dean's Undergraduate Research Fund ($5,000)	2023
COMAP Scholarship ($10,000)	2022
MAA Award in Mathematical Contest in Modeling (Top 0.1%)	2022
Bronze Medal of Shing-Tung Yau Computer Science Award ($10,000)	2019
Finalist of FIRST Robotics Competition World Championship (Top 0.2%)	2019
First Prize of Chinese Mathematical Olympiad (Top 0.1%)	2018

Services

- Conference Reviewer: NeurIPS 2023, ICLR 2024
- Teaching assistant of CSCI-UA 310 Basic Algorithms
- Executive board member of NYU Mathematics Society
- Influential writer on Zhihu forum with 27K followers and 16,000K total reads

Skills

- **Skills:** Python (Pytorch), C/C++, Java, html, css, Linux, LaTeX, Assembly language
- **Languages:** English (TOEFL 114/120), Chinese (Native)